Assignment 1 - Git & GitHub Fundamentals

Session 1 | Group D | Individual Lab Work

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Repository: https://github.com/KBLLL05/git-session1-KBLLL05

1. Repository Setup

The supplied starter HTML and CSS were imported into git-session1-KBLLL05, then git init -b main initialized the repository. The .gitignore excludes local.env, .env, and *.log; !important.log keeps the required log visible. git check-ignore -v verified the rules, and git ls-files confirmed that important.log is tracked. A .gitattributes rule treats PDF files as binary to preserve their bytes across platforms.

2. README & Commit History

The README contains my name, GitHub username, project description, HTML/CSS/Git/Markdown technologies, a shell code block, and links. The AUI logo was extracted from the README example on lecture slide 12 because the starter ZIP did not contain a separate image. The initial work was split into five focused commits: starter setup, README, homepage structure, styling, and the About feature. Later commits record the merge, report preparation, identity/logo updates, and remote synchronization.

3. Selective Staging

index.html and style.css were modified together. git diff showed unstaged edits. git add index.html staged only the homepage; git diff --cached showed only that file. Commit 2dea208 recorded the homepage while git status --short still showed style.css modified. The stylesheet was committed separately as 13a8d16. This demonstrates that a commit records the index, not every change in the working directory.

4. Branch & Merge

git switch -c feature/about created the feature branch. Commit 38d8167 added about.html and a homepage link. After git switch main, git merge --no-ff feature/about created merge commit a637cda. The graph on page 3 shows that the feature branch was merged into main.

5. Remote Changes

An empty public GitHub repository was created, connected using git remote add origin, and published with git push -u origin main. A commit saves a staged snapshot locally; push uploads commits to a remote. The README was edited directly in the GitHub browser editor, creating f36e02a. git fetch origin updated origin/main while leaving local HEAD and the README unchanged. git status -sb showed [behind 1], and git rev-list --left-right --count HEAD...origin/main returned 0 1. git pull --ff-only origin main then integrated that commit; the comparison became 0 0. Fetch updates remote-tracking information; pull fetches and integrates it into the current branch.

6. Fair Use of AI

I used AI assistance while preparing the project and report. My requests focused on the assignment requirements, Git workflow, and possible improvements to the explanations. Suggestions were incorporated and adjusted to match the project details. The resulting work was checked against the assignment instructions and the recorded Git status, diffs, and commit history.

7. Sources & Reflection

Sources: Assignment 1 and Report Template - Group D.pdf (requirements and report headings); git presentation - session 1 - group D.pdf (Git workflow and slide 12 logo); Session1_Practice - group D.zip (starter HTML, CSS, and practice logs). The most subtle concept in this workflow is separating the working tree, staging area, and repository history: a modified file stays outside a commit until it is staged. When Git behaves unexpectedly, the first checks are git status, the current branch, and git diff versus git diff --cached.

Troubleshooting

A controlled git add app.log attempt failed because *.log is ignored. git check-ignore -v app.log identified the exact rule; the file was correctly left untracked instead of forcing it into Git. Network commands also initially failed in the restricted execution environment; after network approval, push, fetch, and pull succeeded without changing repository URLs or ignore rules.

Evidence - recorded Git states

These are excerpts of real command output. Full transcripts are in evidence/local-git-evidence.txt and evidence/remote-git-evidence.txt.

```
$ git status --short  # after staging only index.html
M  index.html
 M style.css

$ git status --short  # after commit 2dea208
 M style.css

$ git check-ignore -v app.log debug.log local.env
.gitignore:3:*.log       app.log
.gitignore:3:*.log       debug.log
.gitignore:1:local.env   local.env

$ git status --short --ignored  # after local feature merge
!! app.log
!! debug.log
!! local.env
```

After fetch, before pull: local HEAD stayed at a58110e. The README blob hash before and after fetch was fb039e60cc268b369921e877c90732a1b7d22874.

```
$ git status -sb
## main...origin/main [behind 1]
$ git rev-list --left-right --count HEAD...origin/main
0  1
$ git log --oneline HEAD..origin/main
f36e02a Document remote synchronization in GitHub editor

$ git pull --ff-only origin main
Updating a58110e..f36e02a
Fast-forward
 README.md | 2 ++
 1 file changed, 2 insertions(+)
$ git rev-list --left-right --count HEAD...origin/main
0  0
```

History through the GitHub README edit (the final report commit follows this snapshot):

```
* f36e02a Document remote synchronization in GitHub editor
* a58110e Add student details and university logo from lecture
* ae18643 Preserve PDF report as binary across platforms
* 79d53e8 Add draft lab report and verified local workflow evidence
*   a637cda Merge About feature into main
|\
| * 38d8167 Add About page and homepage navigation
|/
* 13a8d16 Improve page spacing colors and readability
* 2dea208 Structure homepage and connect stylesheet
* 673d31b Document project purpose technologies and learning goals
* ab536b2 Initialize starter project and ignore local configuration
```

Evidence - README rendered on GitHub

The browser screenshot shows the GitHub commit, rendered Markdown, student name, technologies, code block, university logo, and remote synchronization sentence.

